

DR-2 residual formalised: the cross-energy lemma, and the reduction to the Pencil Rigidity / Pair-Sum Surface Multiplicity conjecture

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

DR-2 is the unrestricted-class sphere additive-energy bound: for every finite $Q \subset S^2$, $E_+(Q) \leq C N^2 \log^B N$, the open route to closing STEP-5B without the H-ADM-COH amended class. It is OPEN and off the critical path (B5 is T5 PINNED-CLOSURE on the amended class; B1 is T6). The operator-supplied external research (Math447–469) reduces DR-2 to a single named conjecture. This note formalises that residual at honest tiers and registers one verified supporting lemma. It changes no tier.

2. The cross-energy lemma (T7 PROVED) and a correction

Lemma. Let $A \subset C_1$, $B \subset C_2$ be finite subsets of circles on S^2 . With $r_P(w) = \#\{(p, p') \in P^2 : p - p' = w\}$, the off-diagonal cross energy $E_{\text{off}}(A, B) := E_+(A, B) - |A||B|$ satisfies

$$E_{\text{off}}(A, B) \leq 2 \min(|A|^2 - |A|, |B|^2 - |B|), \quad E_+(A, B) \leq |A||B| + 2 \min(|A|^2, |B|^2) \leq 3|A||B|.$$

Proof. For $w \neq 0$, a circle and its translate meet in at most two points, so $r_P(w) \leq 2$. Since $E_+(A, B) = \sum_w r_A(w)r_B(-w)$ and the $w = 0$ term is $r_A(0)r_B(0) = |A||B|$,

$$E_{\text{off}}(A, B) = \sum_{w \neq 0} r_A(w)r_B(-w) \leq 2 \sum_{w \neq 0} r_A(w) = 2(|A|^2 - |A|),$$

and symmetrically with B . $\square$

Correction (devil's-advocate, code-discipline). The external Pass 4 states $E_+(A, B) \leq 2|A||B|$ by applying $r \leq 2$ to ALL w , including $w = 0$ where $r_P(0) = |P| \not\leq 2$. This is false for the full energy: the witness $A = B = 16$ equally-spaced great-circle points has $E_+ = 720 > 512 = 2|A||B|$. The corrected bounds above hold: the same witness has $E_{\text{off}} = 464 \leq 480 = 2(|A|^2 - |A|)$. The non-parallel hypothesis of Pass 4 is moreover NOT load-bearing – the $r \leq 2$ mechanism holds for parallel and non-parallel circles alike (verified on all four configurations). The qualitative $E_+(A, B) = O(|A||B|)$ and the reduction below are unaffected; only the constant $2 \rightarrow 3$ (full) changes, and the cluster-decomposition uses E_{off} with constant 2.

3. The Pencil Rigidity / PSM conjecture (T2 CONJECTURE)

Conjecture (Pair-Sum Surface Multiplicity). There exist C, B such that for every finite $Q \subset S^2$ decomposed into its circle-rich part, the non-structured circle-pair interaction obeys

$$\sum_s \left(\sum_{\alpha, \beta} r_{\alpha\beta}(s) \right)^2 \leq C N^2 \log^B N,$$

where $r_{\alpha\beta}(s)$ counts pairs $(a \in C_\alpha, b \in C_\beta)$ with $a + b = s$. Equivalently (Pencil Rigidity): rich circle pencils sharing a common pair-sum surface have $O(\log^B N)$ controlled overlap.

Pre-registered falsification gate. The conjecture (hence DR-2) is **FALSIFIED** by any explicit family $Q_N \subset S^2$ with $E_+(Q_N) \geq N^{2+\delta}$ for a fixed $\delta > 0$ (super-polylogarithmic additive energy). The external research's Pass-27 counterexample audit and the Pass-9 no-small-doubling probe found no such family; the single-circle ($K = 14$, R-002), coaxial ($K \leq 30$), and rectangle ($R = O(n^{5/2})$, R-007) structured classes are all $O(N^2)$ or better, consistent with the conjecture.

4. The reduction $\text{PSM} \Rightarrow \text{DR-2}$ (T3 PROOF SKETCH)

Claim. The PSM conjecture (equivalently Pencil Rigidity) implies DR-2.

Sketch. Decompose $Q = Q_{\text{thin}} \cup Q_{\text{struct}} \cup Q_{\text{rich}}$. (i) The circle-thin part obeys $E_+(Q_{\text{thin}}) \leq L N^2$ with $L = \log^B N$ (few points per circle). (ii) The structured circle-rich part (single circles, coaxial pencils) obeys $E_+(Q_{\text{struct}}) \leq C N^2$ by the closed estimates R-002/R-003/R-004/R-007. (iii) The remaining non-structured circle-rich interaction is exactly the pair-sum surface multiplicity, bounded by $C N^2 \log^B N$ under PSM. (iv) Cross terms between the three parts are bounded by Cauchy-Schwarz, $E_+(X, Y) \leq \sqrt{E_+(X)E_+(Y)}$, and by the cross-energy lemma of §2 for circle pairs. Summation gives $E_+(Q) \leq C' N^2 \log^{B'} N$. $\square$

Marked gaps (the T3 obstructions). OPEN GAP α : the existence of the controlled decomposition into $O(\log^B N)$ structured clusters + a thin remainder is exactly the PSM/Pencil Rigidity conjecture (not independently proved). OPEN GAP β : the uniformity of the per-cluster constants C_α across the dyadic richness scales (Pass-21) is asserted, not proved. These are the residual obstructions that keep DR-2 at T1 OPEN and the reduction at T3.

5. Status and relation to the repository

DR-2 stays OPEN, off the critical path. This note (i) corrects and registers the cross-energy lemma (T7, R-021), (ii) registers the PSM/Pencil Rigidity conjecture (T2) with a falsification gate, and (iii) records the reduction $\text{PSM} \Rightarrow \text{DR-2}$ (T3). The DR2-SHARE residual is thereby sharpened from the loose “carrier-richness $\chi(P) = O(\text{polylog})$ ” framing to a single named additive-combinatorics conjecture. No claim tier moves: B5 stays T5, B1 stays T6, and the H-ADM-COH amended-class closure remains the official STEP-5B route. The note is the seed for the standalone paper “Additive Energy and Spherical Rectangle Counts on S^2 ”.

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

(α) “Is the cross-energy lemma over-claimed at T7?” **DISMISSED**: the proof is elementary and unconditional ($r \leq 2$ from circle $\cap$ translate ≤ 2), and is numerically confirmed on four

configurations including the structured witness; it is T7 in the same sense as R-002/R-003/R-004. The external constant error was caught and corrected here.

- (β) “Is registering the conjecture as T2 an over-claim?” DISMISSED: T2 requires only a precise statement, partial evidence, and a pre-registered falsification gate – all present (§3). It is explicitly a CONJECTURE, not a theorem; the reduction (§4) is T3 with marked gaps.
- (γ) “Does any of this move a claim tier?” DISMISSED: DR-2 is OPEN, off critical path; B5 stays T5, B1 stays T6. The note formalises a residual and registers one supporting lemma; it performs no tier action. The external $T7_{\text{conditional-robust}}$ label is NOT adopted.

Quantitative sanity check (mandatory): magnitude – the witness $E_+ = 720$, $E_{\text{off}} = 464 \leq 480$ (within 3.3% of the ceiling), $2|A||B| = 512$ violated (the correction); sign-direction – all energies and bounds positive; limit-case – generic non-coincident circle pairs give $E_+ = |A||B|$ (diagonal only), well under $3|A||B|$; consistency – $r_{\text{max}} \leq 2$ on every circle, parallel and non-parallel, matching the geometric mechanism.

7. Result footer

Result ID:	B5 / DR-2 residual formalisation (cross-energy lemma + PSM conjecture + reduction)
Precise statement:	(a) T7: $E_{\text{off}}(A,B) \leq 2 \min(A ^2 - A , B ^2 - B)$, $E_+(A,B) \leq 3 A B $ for A,B on sphere-circles ($r \leq 2$ for $w \neq 0$); corrects the external $2 A B $ (off by the $w=0$ diagonal; witness $E_+=720 > 512$). (b) T2: PSM/Pencil Rigidity conjecture $\sum_s (\sum r_{\text{ab}}(s))^2 \leq C N^2 \log^B N$ + falsification gate $E_+(Q_N) \geq N^{2+\delta}$. (c) T3: PSM $\Rightarrow$ DR-2 (decomposition + Cauchy-Schwarz; gaps α/β marked).
Scope:	DR-2 research branch, off critical path. Sphere-circle additive energy.
Dependencies:	R-002/R-003/R-004/R-007 (closed structured estimates); external research Math447-469 (operator-supplied); strategy/dr2-external-research-assessment-260608.md.
Evidence grade:	(a) EXECUTED (dr2_cross_energy_lemma.py 6/6) + analytic; (b) CONJECTURE; (c) PROOF SKETCH.
Reproduction command:	python codes/vacuum/dr2_cross_energy_lemma.py
Expected output:	6/6 PASS; $r_{\text{max}} \leq 2$; $E_{\text{off}} \leq 2 \min(n^2 - n)$; witness $E_+ = 720$ falsifies $2 A B = 512$; $E_+ \leq 3 A B $.
Falsification gate:	An explicit Q_N subset S^2 with $E_+(Q_N) \geq N^{2+\delta}$ ($\delta > 0$) falsifies PSM and DR-2.
Tier before / after:	DR-2: OPEN $\rightarrow$ OPEN (formalised; reduced to PSM/Pencil Rigidity). B5: T5 $\rightarrow$ T5. B1: T6 $\rightarrow$ T6. No tier action.
No-overclaim statement:	DR-2 is NOT closed. The cross-energy lemma is T7; the conjecture is T2; the reduction is T3 with marked gaps. The external $T7_{\text{conditional-robust}}$ label is not adopted; the external $2 A B $ constant is corrected.
Next required action:	Prove or disprove the PSM / Pencil Rigidity conjecture (independent math branch); none required for the TECT mainline (H-ADM-COH route stands).